Math 675 Spring 2026:
Homework 3 (LATE)
Due Tuesday, February 10th, 2026

Question 1:

a) $\mathbf{x}' = \begin{pmatrix} \frac{5}{4} & \frac{3}{4} \\ \frac{3}{4} & \frac{5}{4} \end{pmatrix} \mathbf{x}$

Finding the general Solution

$$\begin{vmatrix} \frac{5}{4} - \lambda & \frac{3}{4} \\ \frac{3}{4} & \frac{5}{4} - \lambda \end{vmatrix} = \left(\tfrac{5}{4} - \lambda\right)^2 - \left(\tfrac{3}{4}\right)^2 = 0$$

$$= \left(\tfrac{5}{4} - \lambda + \tfrac{3}{4}\right)\left(\tfrac{5}{4} - \lambda - \tfrac{3}{4}\right) = 0$$

$$= (2 - \lambda)\left(\tfrac{1}{2} - \lambda\right) = 0 \Rightarrow \lambda_1 = 2, \lambda_2 = \tfrac{1}{2}$$

Eigenvalue #1

$$\lambda_{\mathbf{1}} = \quad \mathbf{2} : \begin{bmatrix} \frac{5}{4} - \lambda_1 & \frac{3}{4} \\ \frac{3}{4} & \frac{5}{4} - \lambda_1 \end{bmatrix} \to \begin{bmatrix} -\frac{3}{4} & \frac{3}{4} \\ \frac{3}{4} & -\frac{3}{4} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} -1v_1 + 1v_2 &= 0 \\ 1v_1 - 1v_2 &= 0 \end{aligned} \longrightarrow \quad v_1 = v_2 \quad \longrightarrow \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Eigenvalue #2

$$\lambda_{\mathbf{2}} = \quad \frac{\mathbf{1}}{\mathbf{2}} : \begin{bmatrix} \frac{5}{4} - \lambda_2 & \frac{3}{4} \\ \frac{3}{4} & \frac{5}{4} - \lambda_2 \end{bmatrix} \to \begin{bmatrix} \frac{3}{4} & \frac{3}{4} \\ \frac{3}{4} & \frac{3}{4} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} 1v_1 + 1v_2 &= 0 \\ 1v_1 + 1v_2 &= 0 \end{aligned} \longrightarrow \quad v_1 = -v_2 \quad \longrightarrow \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

So our general solution is $\mathbf{x} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{\frac{1}{2}t}$

Describing the behavior of the system

As $t \to \infty$, we can see that e^{2t} approaches infinity significantly faster than $e^{0.5t}$,so solutions will become parrellel to the line $y = x$ as $t \to \infty$.

$$\lim_{t\to\infty} \frac{c_1 e^{2t} v_1}{c_2 e^{0.5t} v_2} = \lim_{t\to\infty} \frac{c_1 v_1}{c_2 v_2} e^{1.5t} = \infty$$

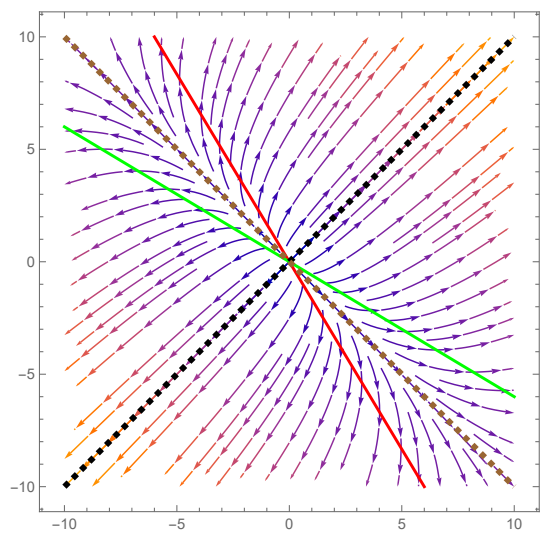
Sketching the phase portrait

We want to find the nullclines of the system, so we set each derivative to 0:

Set $x' = 0$: so $0 = \frac{5}{4}x + \frac{3}{4}y$, giving us $y = -\frac{5}{3}x$
Set $y' = 0$: so $0 = \frac{3}{4}x + \frac{5}{4}y$, giving us $y = -\frac{3}{5}x$

We still also have our initial eigenvalue directions of $y = x$ and $y = -x$

Using technology to draw the phase portrait



- The black dashed line is the fast eigendirection $y = x$, which is the direction that trajectories will become parallel to as $t \to \infty$.
- The brown dashed line is the slow eigendirection $y = -x$. Trajectories emerge from the origin tangent to this line as $t \to -\infty$.
- The x-nullcline in red is $y = -\frac{5}{3}x$. Trajectories cross vertically.
- The y-nullcline in green is $y = -\frac{3}{5}x$. Trajectories cross horizontally.
- Trajectories between the nullclines spread outwards from the brown eigendirection, and trajectories outside of the nullclines spread away to be parallel to the black eigendirection.

b) $\mathbf{x}' = \begin{pmatrix} 4 & -3 \\ 8 & -6 \end{pmatrix} \mathbf{x}$

Finding the general Solution

$$\begin{vmatrix} 4-\lambda & -3 \\ 8 & -6-\lambda \end{vmatrix} = (4-\lambda)(-6-\lambda) - (-3)(8) = 0$$

$$= -24 - 4\lambda + 6\lambda + \lambda^2 + 24 = 0$$

$$= \lambda^2 + 2\lambda = 0$$

$$\Rightarrow \lambda_1 = 0, \lambda_2 = -2$$

Eigenvalue #1

$$\lambda_{\mathbf{1}} = \quad \mathbf{0} : \begin{bmatrix} 4-\lambda_1 & -3 \\ 8 & -6-\lambda_1 \end{bmatrix} \rightarrow \begin{bmatrix} 4 & -3 \\ 8 & -6 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} 4v_1 - 3v_2 &= 0 \\ 8v_1 - 6v_2 &= 0 \end{aligned} \longrightarrow \quad 4v_1 = 3v_2 \quad \longrightarrow \mathbf{v}_1 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Eigenvalue #2

$$\lambda_{\mathbf{2}} = -\mathbf{2} : \begin{bmatrix} 4-\lambda_2 & -3 \\ 8 & -6-\lambda_2 \end{bmatrix} \rightarrow \begin{bmatrix} 6 & -3 \\ 8 & -4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} 6v_1 - 3v_2 &= 0 \\ 8v_1 - 4v_2 &= 0 \end{aligned} \longrightarrow \quad 2v_1 = v_2 \quad \longrightarrow \mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

So our general solution is $\mathbf{x} = c_1 \begin{pmatrix} 3 \\ 4 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{-2t}$

Describing the behavior of the system

As $t \to \infty$, we can see that ther single term e^{-2t} approaches zero. But the first term does not change at all,so solutions will become parrellel to the line $y = \frac{1}{2}x$ as $t \to \infty$.

$$\lim_{t\to\infty} \frac{c_2 e^{-2t} v_2}{c_1 v_1} = 0$$

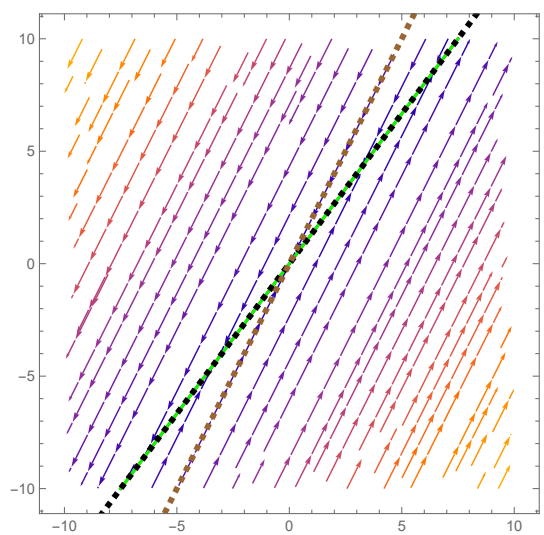
Sketching the phase portrait

We want to find the nullclines of the system, so we set each derivative to 0:

Set $x' = 0$: so $0 = 4x - 3y$, giving us $y = \frac{4}{3}x$
Set $y' = 0$: so $0 = 8x - 6y$, giving us $y = \frac{4}{3}x$

We still also have our initial eigenvalue directions of $y = \frac{4}{3}x$ and $y = 2x$

Using technology to draw the phase portrait



- The brown dashed line is the eigendirection $y = 2x$, which is the direction that trajectories will become parallel to as $t \to \infty$.
- The black dashed line is the eigendirection $y = \frac{4}{3}x$.
  Trajectories are just unmoving singular points on this line.
- The green line are the $x-$ and y-nullclines $y = \frac{4}{3}x$.
  Trajectories don't cross but stop at this line
- This means that every point along the green/black lines are equlibrium points, and trajectories that start off the line will approach the line

c) $\mathbf{x}' = \begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix} \mathbf{x}$

Finding the general Solution

$$\begin{vmatrix} 3-\lambda & -2 \\ 4 & -1-\lambda \end{vmatrix} = (3-\lambda)(-1-\lambda) - (-2)(4) = 0$$

$$= -3 - 3\lambda + \lambda + \lambda^2 + 8 = 0$$

$$= \lambda^2 - 2\lambda + 5 = 0$$

$$\Rightarrow \lambda_1 = 1 + 2i, \lambda_2 = 1 - 2i$$

Eigenvalue #1

$$\lambda_{\mathbf{1}} = \mathbf{1} + \mathbf{2i} : \begin{bmatrix} 3-\lambda_1 & -2 \\ 4 & -1-\lambda_1 \end{bmatrix} \to \begin{bmatrix} 2-2i & -2 \\ 4 & -2-2i \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} (2-2i)v_1 - 2v_2 &= 0 \\ 4v_1 - (2+2i)v_2 &= 0 \end{aligned} \longrightarrow \quad v_2 = (1-i)v_1 \quad \longrightarrow \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1-i \end{bmatrix}$$

So our general solution is $\mathbf{x} = c_1 \begin{pmatrix} 1 \\ 1-i \end{pmatrix} e^{(1+2i)t}$

This just looks bad and its hard to interpret, so let's use Euler's formula to rewrite it in a better way:

$$\begin{pmatrix} 1 \\ 1-i \end{pmatrix} e^{(1+2i)t} = \begin{pmatrix} 1 \\ 1-i \end{pmatrix} e^t e^{2it} = \begin{pmatrix} 1 \\ 1-i \end{pmatrix} e^t(\cos(2t) + i\sin(2t))$$

$$= e^t \left[\begin{pmatrix} \cos(2t) + i\sin(2t) \\ (1-i)(\cos(2t) + i\sin(2t)) \end{pmatrix} \right]$$

$$= e^t \left[\begin{pmatrix} \cos(2t) \\ \cos(2t) + \sin(2t) \end{pmatrix} + i \begin{pmatrix} \sin(2t) \\ \sin(2t) - \cos(2t) \end{pmatrix} \right]$$

$$= c_1 e^t \left[\begin{pmatrix} \cos(2t) \\ \cos(2t) + \sin(2t) \end{pmatrix} \right] + c_2 e^t \left[\begin{pmatrix} \sin(2t) \\ \sin(2t) - \cos(2t) \end{pmatrix} \right]$$

$$= c_1 e^t \left[\begin{pmatrix} 1 \\ 1 \end{pmatrix} \cos 2t - \begin{pmatrix} 0 \\ -1 \end{pmatrix} \sin 2t \right] + c_2 e^t \left[\begin{pmatrix} 0 \\ -1 \end{pmatrix} \cos 2t + \begin{pmatrix} 1 \\ 1 \end{pmatrix} \sin 2t \right]$$

Notice that the vectors in front of the sins and cosines are components of the eigenvector that we started with:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1-i \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + i \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

Since going through all of that work with complex eigenvalues, I wonder if we could build a potential shortcut? If $\lambda = \mathbf{a} + \mathbf{b}i$ is an eigenvalue of A with eigenvector $\mathbf{v} = \mathbf{c} + i\mathbf{d}$, then we maybe can write the solution as

$$\mathbf{x}(t) = e^{\mathbf{a}t}(\mathbf{c}\cos(\mathbf{b}t) - \mathbf{d}\sin(\mathbf{b}t)) + e^{\mathbf{a}t}(\mathbf{d}\sin(\mathbf{b}t) + \mathbf{c}\cos(\mathbf{b}t))$$

Describing the behavior of the system

As $t \to \infty$, we can see that both terms approach infinity. But due to the sins and cosines, and the nature of complex solutions, all trajectories will be spiraling outwards

One thing I'm not quite sure of, maybe I missed it from the notes, but finding out the angle or tilt direction would be even more useful to describe the behavior of the system and to be able to better sketch it by hand.

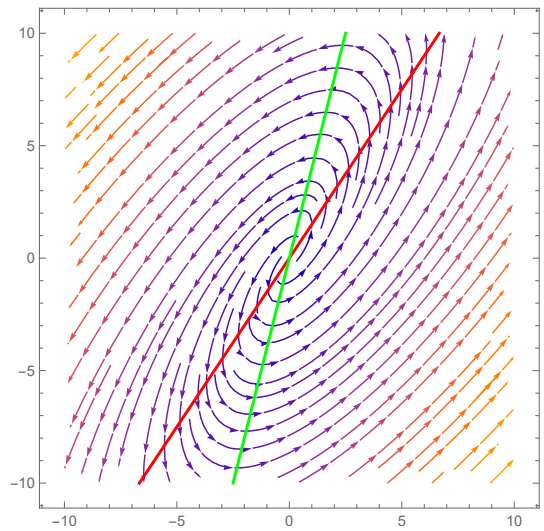
Sketching the phase portrait

We want to find the nullclines of the system, so we set each derivative to 0:

Set $x' = 0$: so $0 = 3x - 2y$, giving us $y = \frac{3}{2}x$
Set $y' = 0$: so $0 = 4x - 1y$, giving us $y = 4x$

We still also have our initial eigenvalue directions but we can't plot them since they are complex

Using technology to draw the phase portrait



- The eigenvalues are complex ($\lambda = 1 \pm 2i$). Since the real part is positive, the origin is an unstable spiral source.
- There are no real eigendirections for this system; instead, trajectories spiral outward from the origin.
- The red line is the x-nullcline $y = \frac{3}{2}x$.
  Trajectories cross this line vertically.
- The green line is the y-nullcline $y = 4x$.
  Trajectories cross this line horizontally.

d) $\mathbf{x}' = \begin{pmatrix} 2 & -\frac{5}{2} \\ \frac{9}{5} & -1 \end{pmatrix} \mathbf{x}$

Finding the general Solution

$$\begin{vmatrix} 2-\lambda & -\frac{5}{2} \\ \frac{9}{5} & -1-\lambda \end{vmatrix} = (2-\lambda)(-1-\lambda) - (-\tfrac{5}{2})(\tfrac{9}{5}) = 0$$

$$= -2 - 2\lambda + \lambda + \lambda^2 + \tfrac{9}{2} = 0$$

$$= \lambda^2 - \lambda + \tfrac{5}{2} = 0$$

$$\Rightarrow \lambda_1 = \tfrac{1}{2} + \tfrac{3}{2}i, \lambda_2 = \tfrac{1}{2} - \tfrac{3}{2}i$$

Eigenvalue #1

$$\lambda_{\mathbf{1}} = \mathbf{\frac{1}{2}} + \mathbf{\frac{3}{2}i} : \begin{bmatrix} 2-\lambda_1 & -\frac{5}{2} \\ \frac{9}{5} & -1-\lambda_1 \end{bmatrix} \to \begin{bmatrix} \frac{3}{2} - \frac{3}{2}i & -\frac{5}{2} \\ \frac{9}{5} & -\frac{3}{2} - \frac{3}{2}i \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} (\tfrac{3}{2} - \tfrac{3}{2}i)v_1 - \tfrac{5}{2}v_2 &= 0 \\ \tfrac{9}{5}v_1 - (-\tfrac{3}{2} + \tfrac{3}{2}i)v_2 &= 0 \end{aligned} \to \quad (3-3i)v_1 = 5v_2 \quad \to \mathbf{v}_1 = \begin{bmatrix} 5 \\ 3-3i \end{bmatrix}$$

So our general solution is $\mathbf{x} = c_1 \begin{pmatrix} 5 \\ 3-3i \end{pmatrix} e^{(\frac{1}{2}+\frac{3}{2}i)t}$

This just looks bad and its hard to interpret, so let's use Euler's formula to rewrite it in a better way:

$$\begin{pmatrix} 5 \\ 3-3i \end{pmatrix} e^{(\frac{1}{2}+\frac{3}{2}i)t} = \begin{pmatrix} 5 \\ 3-3i \end{pmatrix} e^{\frac{1}{2}t} e^{\frac{3}{2}it} = \begin{pmatrix} 5 \\ 3-3i \end{pmatrix} e^t(\cos(\tfrac{3}{2}t) + i\sin(\tfrac{3}{2}t))$$

$$= e^t \left[\begin{pmatrix} 5(\cos(\frac{3}{2}t) + i\sin(\frac{3}{2}t)) \\ (3-3i)(\cos(\frac{3}{2}t) + i\sin(\frac{3}{2}t)) \end{pmatrix} \right]$$

$$= e^t \left[\begin{pmatrix} 5\cos(\frac{3}{2}t) \\ 3\cos(\frac{3}{2}t) + 3\sin(\frac{3}{2}t) \end{pmatrix} + i \begin{pmatrix} 5\sin(\frac{3}{2}t) \\ 3\sin(\frac{3}{2}t) - 3\cos(\frac{3}{2}t) \end{pmatrix} \right]$$

$$= c_1 e^t \left[\begin{pmatrix} 5\cos(\frac{3}{2}t) \\ 3\cos(\frac{3}{2}t) + 3\sin(\frac{3}{2}t) \end{pmatrix} \right] + c_2 e^t \left[\begin{pmatrix} 5\sin(\frac{3}{2}t) \\ 3\sin(\frac{3}{2}t) - 3\cos(\frac{3}{2}t) \end{pmatrix} \right]$$

$$= c_1 e^t \left[\begin{pmatrix} 5 \\ 3 \end{pmatrix} \cos(\tfrac{3}{2}t) - \begin{pmatrix} 0 \\ -3 \end{pmatrix} \sin(\tfrac{3}{2}t) \right] + c_2 e^t \left[\begin{pmatrix} 0 \\ -3 \end{pmatrix} \cos(\tfrac{3}{2}t) + \begin{pmatrix} 5 \\ 3 \end{pmatrix} \sin(\tfrac{3}{2}t) \right]$$

Notice that the vectors in front of the sins and cosines are components of the eigenvector that we started with:

$$\mathbf{v}_1 = \begin{bmatrix} 5 \\ 3 - 3i \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix} + i \begin{bmatrix} 0 \\ -3 \end{bmatrix}$$

Notice that with another example, our hypothesized shortcut still works! This is pretty cool, so maybe we can just use this formula for any complex eigenvalues we get in the future and save ourselves a lot of time!

Describing the behavior of the system

As $t \to \infty$, we can see that both terms approach infinity. But due to the sins and cosines, and the nature of complex solutions, all trajectories will be spiraling outwards

One thing I'm not quite sure of, maybe I missed it from the notes, but finding out the angle or tilt direction would be even more useful to describe the behavior of the system and to be able to better sketch it by hand.

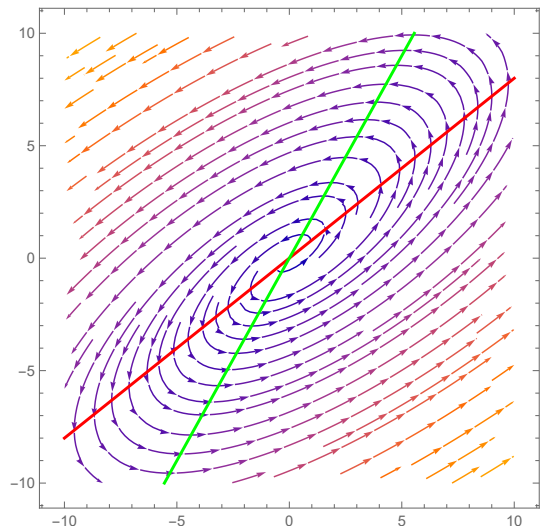
Sketching the phase portrait

We want to find the nullclines of the system, so we set each derivative to 0:

Set $x' = 0$: so $0 = 2x - \frac{5}{2}y$, giving us $y = \frac{4}{5}x$
Set $y' = 0$: so $0 = \frac{9}{5}x - 1y$, giving us $y = \frac{9}{5}x$

We still also have our initial eigenvalue directions but we can't plot them since they are complex

Using technology to draw the phase portrait



- The eigenvalues are complex ($\lambda = \frac{1}{2} \pm \frac{3}{2}i$). Since the real part is positive, the origin is an unstable spiral source.
- There are no real eigendirections for this system; instead, trajectories spiral outward from the origin.
- The red line is the x-nullcline $y = \frac{4}{5}x$.
  Trajectories cross this line vertically.
- The green line is the y-nullcline $y = \frac{9}{5}x$.
  Trajectories cross this line horizontally.

Question 2:

Find the general solution of the 3D System:

$$\mathbf{x}'(t) = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{pmatrix} \mathbf{x}(t)$$

$$\begin{vmatrix} 1-\lambda & 1 & 2 \\ 1 & 2-\lambda & 1 \\ 2 & 1 & 1-\lambda \end{vmatrix} = (1-\lambda)\begin{vmatrix} 2-\lambda & 1 \\ 1 & 1-\lambda \end{vmatrix} - 1\begin{vmatrix} 1 & 1 \\ 2 & 1-\lambda \end{vmatrix} + 2\begin{vmatrix} 1 & 2-\lambda \\ 2 & 1 \end{vmatrix}$$

$$= (1-\lambda)\left[(2-\lambda)(1-\lambda)-1\right] - 1[(1-\lambda)-2] + 2[1-2(2-\lambda)] = 0$$

$$= (1-\lambda)(1-3\lambda+\lambda^2) + (1+\lambda) + (-6+4\lambda) = 0$$

$$= 1 - 3\lambda + \lambda^2 - \lambda + 3\lambda^2 - \lambda^3 - 5 + 5\lambda = 0$$

$$= -\lambda^3 + 4\lambda^2 + \lambda - 4 = 0$$

$$= \lambda^3 - 4\lambda^2 - \lambda + 4 = 0$$

$$= (\lambda^3 - 4\lambda^2) - (\lambda - 4) = 0$$

$$= \lambda^2(\lambda - 4) - (\lambda - 4) = 0$$

$$= (\lambda^2 - 1)(\lambda - 4) = 0$$

So our eigenvalues are: $\lambda_1 = 4, \lambda_2 = 1, \lambda_3 = -1$

Eigenvalue #1

$$\lambda_{\mathbf{1}} = \quad \mathbf{4}: \begin{bmatrix} 1-\lambda_1 & 1 & 2 \\ 1 & 2-\lambda_1 & 1 \\ 2 & 1 & 1-\lambda_1 \end{bmatrix} = \begin{bmatrix} -3 & 1 & 2 \\ 1 & -2 & 1 \\ 2 & 1 & -3 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{This gives us that } \begin{cases} a &= c \\ b &= c \\ c &= c \end{cases} \text{ so our eigenvalue is } \mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Eigenvalue #2

$$\lambda_{\mathbf{2}} = \quad \mathbf{1}: \begin{bmatrix} 1-\lambda_2 & 1 & 2 \\ 1 & 2-\lambda_2 & 1 \\ 2 & 1 & 1-\lambda_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \\ 2 & 1 & 0 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{This gives us that } \begin{cases} a &= c \\ b &= -2c \\ c &= c \end{cases} \text{ so our eigenvalue is } \mathbf{u}_2 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

Eigenvalue #3

$$\lambda_{\mathbf{3}} = -\mathbf{1}: \begin{bmatrix} 1-\lambda_3 & 1 & 2 \\ 1 & 2-\lambda_3 & 1 \\ 2 & 1 & 1-\lambda_3 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 2 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{This gives us that } \begin{cases} a &= -c \\ b &= 0 \\ c &= c \end{cases} \text{ so our eigenvalue is } \mathbf{u}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Thus, the general solution to our system is

$$\mathbf{x}(t) = c_1 e^{4t} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + c_2 e^{t} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} + c_3 e^{-t} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Question 3:

Consider the equation $ay'' + by' + cy = 0$ with a, b, c are constants and $a \neq 0$. The general solution depends on the roots of the characteristic equation ar^2+br+c

a) Given $x_1 = y, x_2 = y'$, we can then also say that $x_1' = y' = x_2$ and $x_2' = y''$ so we can rewrite our original equation as $ax_2' + bx_2 + cx_1 = 0$
Solving for the derivative we obtain $x_2' = -\dfrac{b}{a}x_2 - \dfrac{c}{a}x_1$ with $x_1' = x_2$
Thus, we can write our system as $\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -\frac{c}{a} & -\frac{b}{a} \end{pmatrix} \mathbf{x}$

b) Now let's find the eigenvalues of this system:
$$\begin{vmatrix} -\lambda & 1 \\ -\frac{c}{a} & -\frac{b}{a} - \lambda \end{vmatrix} = (-\lambda)(-\frac{b}{a} - \lambda) + \frac{c}{a} = 0$$
$$= \lambda^2 + \frac{b}{a}\lambda + \frac{c}{a} = a\lambda^2 + b\lambda + c = 0$$
Notice that this is the same characteristic equation as our original second order ODE, so the eigenvalues are just the roots of the characteristic equation.

Question 4:

Consider the system:

$$\mathbf{x}' = \begin{pmatrix} -1 & -1 \\ -\alpha & -1 \end{pmatrix} \mathbf{x}$$

a) Solve the system for $\alpha = 0.5$ What are the eigenvalues? Classify the equilibrium at the origin.

$$\begin{vmatrix} -1-\lambda & -1 \\ -0.5 & -1-\lambda \end{vmatrix} = (-1-\lambda)^2 - (-1)(-0.5) = 0$$

$$= 1 + 2\lambda + \lambda^2 - 0.5 = 0$$

$$= \lambda^2 + 2\lambda + 0.5 = 0$$

$$\Rightarrow \lambda_1 = -1 + \tfrac{\sqrt{2}}{2}, \lambda_2 = -1 - \tfrac{\sqrt{2}}{2}$$

Eigenvalue #1

$$\boldsymbol{\lambda_1 = -1 + \tfrac{\sqrt{2}}{2}} : \begin{bmatrix} -1-\lambda_1 & -1 \\ -0.5 & -1-\lambda_1 \end{bmatrix} \to \begin{bmatrix} -\frac{\sqrt{2}}{2} & -1 \\ -0.5 & -\frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} -\tfrac{1}{\sqrt{2}}v_1 - 1v_2 &= 0 \\ -0.5v_1 - \tfrac{1}{\sqrt{2}}v_2 &= 0 \end{aligned} \longrightarrow \quad v_1 = -\sqrt{2}v_2 \quad \longrightarrow \mathbf{v}_1 = \begin{bmatrix} -\sqrt{2} \\ 1 \end{bmatrix}$$

Eigenvalue #2

$$\boldsymbol{\lambda_2 = -1 - \tfrac{\sqrt{2}}{2}} : \begin{bmatrix} -1-\lambda_2 & -1 \\ -0.5 & -1-\lambda_2 \end{bmatrix} \to \begin{bmatrix} \frac{\sqrt{2}}{2} & -1 \\ -0.5 & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} \tfrac{1}{\sqrt{2}}v_1 - 1v_2 &= 0 \\ -0.5v_1 + \tfrac{1}{\sqrt{2}}v_2 &= 0 \end{aligned} \longrightarrow \quad v_1 = \sqrt{2}v_2 \quad \longrightarrow \mathbf{v}_1 = \begin{bmatrix} \sqrt{2} \\ 1 \end{bmatrix}$$

So our general solution is $\mathbf{x} = c_1 \begin{pmatrix} -\sqrt{2} \\ 1 \end{pmatrix} e^{(-1+\frac{\sqrt{2}}{2})t} + c_2 \begin{pmatrix} \sqrt{2} \\ 1 \end{pmatrix} e^{(-1-\frac{\sqrt{2}}{2})t}$

The equlibrium at the origin is a stable sink node, since both eigenvalues are negative and real.

b) Repeat for $\alpha = 2$

$$\begin{vmatrix} -1-\lambda & -1 \\ -2 & -1-\lambda \end{vmatrix} = (-1-\lambda)^2 - (-1)(-2) = 0$$

$$= 1 + 2\lambda + \lambda^2 - 2 = 0$$

$$= \lambda^2 + 2\lambda - 1 = 0$$

$$\Rightarrow \lambda_1 = -1 + \sqrt{2}, \lambda_2 = -1 - \sqrt{2}$$

Eigenvalue #1

$$\boldsymbol{\lambda_1 = -1 + \sqrt{2}} : \begin{bmatrix} -1-\lambda_1 & -1 \\ -2 & -1-\lambda_1 \end{bmatrix} \to \begin{bmatrix} -\sqrt{2} & -1 \\ -2 & -\sqrt{2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} -\sqrt{2}v_1 - 1v_2 &= 0 \\ -2v_1 - \sqrt{2}v_2 &= 0 \end{aligned} \longrightarrow \quad v_2 = -\sqrt{2}v_1 \quad \longrightarrow \mathbf{v}_1 = \begin{bmatrix} 1 \\ -\sqrt{2} \end{bmatrix}$$

Eigenvalue #2

$$\boldsymbol{\lambda_2 = -1 - \sqrt{2}} : \begin{bmatrix} -1-\lambda_2 & -1 \\ -2 & -1-\lambda_2 \end{bmatrix} \to \begin{bmatrix} \sqrt{2} & -1 \\ -2 & \sqrt{2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} \sqrt{2}v_1 - 1v_2 &= 0 \\ -2v_1 + \sqrt{2}v_2 &= 0 \end{aligned} \longrightarrow \quad v_2 = \sqrt{2}v_1 \quad \longrightarrow \mathbf{v}_1 = \begin{bmatrix} 1 \\ \sqrt{2} \end{bmatrix}$$

So our general solution is $\mathbf{x} = c_1 \begin{pmatrix} 1 \\ -\sqrt{2} \end{pmatrix} e^{(-1+\sqrt{2})t} + c_2 \begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix} e^{(-1-\sqrt{2})t}$

The equlibrium at the origin is a unstable saddle, since we have one positive and one negative real eigenvalue.

c) Find the eigenvalues in terms of α and determine the value of α between 0.5 and 2 where the transition from one type of behavior to the other occurs.

$$\begin{vmatrix} -1-\lambda & -1 \\ -\alpha & -1-\lambda \end{vmatrix} = (-1-\lambda)^2 - \alpha = 0$$

$$\rightarrow (-1-\lambda)^2 = \alpha$$

$$\rightarrow -1-\lambda = \pm\sqrt{\alpha}$$

$$\rightarrow -1-\lambda = \pm\sqrt{\alpha}$$

$$\rightarrow \lambda_1 = -1+\sqrt{\alpha}, \lambda_2 = -1-\sqrt{\alpha}$$

We can see that $\lambda_1 > \lambda_2$ for all $\alpha > 0$. Notice that for all values of α, λ_2 is always negitive, but the same cant be said for λ_1. So the transition from stable to unstable behavior occurs when λ_1 goes from negative to positive, which happens at $\alpha = 1$. Any values of α between 0 and 1 will give us both negitive eigenvalues, meaning the origin is a stable sink node, and any values of α between 1 and 2 will give us an unstable saddle since we will obtain one positive and one negative eigenvalue.